

Performance Analysis of Free Space Optical Communication in Urban Rain Environment

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Abstract— This study investigates the feasibility of Free Space Optical (FSO) communication systems for high-speed data transmission, specifically in the context of Dhaka, Bangladesh. FSO systems, which use modulated light beams, offer an alternative to fiber optics by providing cost-effective deployment, high bandwidth, and secure data transfer. However, atmospheric conditions like rain, fog, and dust introduce attenuation and scattering, impacting system performance. Utilizing rainfall data collected from the CHRS Data Portal for Dhaka, this study analyzes rain attenuation's effects on the signal-to-noise ratio (SNR), bit-error rate (BER), and channel capacity under varying conditions. The findings highlight FSO communication's viability for urban environments, provided suitable mitigation strategies are employed.

Keywords— FSO, optical communication, Free space optical, Laser communication, Optical Wireless Technology, Urban Communication.

I. INTRODUCTION

Free-space optical (FSO) communication is a line-of-sight technology that transmits data via modulated light beams through the atmosphere, offering an alternative to fiber optics. FSO uses air for transmission, while fiber relies on enclosed glass cables. FSO's main advantage is its cost-effective deployment through line-of-sight communication. Existing wireless systems often overcrowd radio and microwave bands, making FSO's massive bandwidth an appealing solution for bandwidth consumption [1],[2]. FSO technology uses visible to near-infrared light propagating through the atmosphere to transmit information. The FSO communications are attracting attention as the contemporary engineering science to resolve the last mile bottleneck issues in local area access networks due to their high bandwidth, low-cost implementation in a non-licensed spectrum, relatively low power consumption, and immunity and security compared with RF technologies [3]. The main

challenge in the FSO system is using laser beams transmitted through the atmosphere, as the attenuation is insatiable and challenging to predict. The wavelength of atmospheric composites such as rain, fog, dust, and smoke particles is similar to the optical wavelength used in FSO, which causes a reduction in performance and lowers the signal quality status by scattering and absorption of visible optical beams (NIR). However, in solution, we can analyze historical weather data from the airport to forecast the FSO system's link availability as a function of distance for different geometric areas, and these availability curves help to determine the practical range for the FSO system.

FSO addresses many applications like metropolitan networks, inter-building communication, back-haul wireless systems, indoor links, fiber backup, service acceleration, security, military purposes, satellite communication, etc. [1], [2], [4], [5]. Some laboratory demonstrations show the ability to transmit massive data rates reaching the terabit. Commercial products are available for an on-off-keying (OOK) modulated data bandwidth of 100 Mbps up to 10 Gbps, depending on the length of the link [6]. Although there are many advantages and various FSO applications, the FSO link is affected by weather conditions such as fog, rain, dust, snow, turbulence, etc. Most of the literature considers the effect of turbulence that occurs because of air heating by the Sun. FSO channel models under this impairment have been proposed for single-input single-output (SISO) wireless systems and for multiple-input multiple-output (MIMO) systems [8]. Moreover, some techniques, such as relays, are proposed to improve the system's performance [8].

Rain poses a significant challenge for free-space optics (FSO) systems, causing attenuation of hundreds of decibels per kilometer. This high attenuation may lead to link outages and reduced availability. Effective rain modeling is essential to enhance reliability and reduce download times. We assess